2025 to 2050

Future-network attribute transfer

Two tested methods for calibrated QVDF continuity

TESTED ON 49,662 LINKS

Prepared from the NVTA 2025 calibrated network and 2050 No-Build true-shape network

The node-pair problem is concentrated in 428 links

Most of the future network remains exactly traceable to the calibrated 2025 network

49,662
directed links in the 2050 network

49,234
exact calibrated node-pair matches

428
future pairs missing from calibration

0.862%
of future links require a policy



102 calibrated 2025 pairs no longer exist in 2050

+220 net increase in directed network links

Method 1 is deterministic and safe

Exact calibrated pairs are copied. Every other link receives QVDF values from its link type

1

Exact pair

Use the calibrated 2025 AM, MD, and PM QVDF values

2

Missing pair

Write the directed pair and link context to the audit log

3

QVDF fallback

Use ATYPE times 100 + FTYPE, with the all-row fallback

4

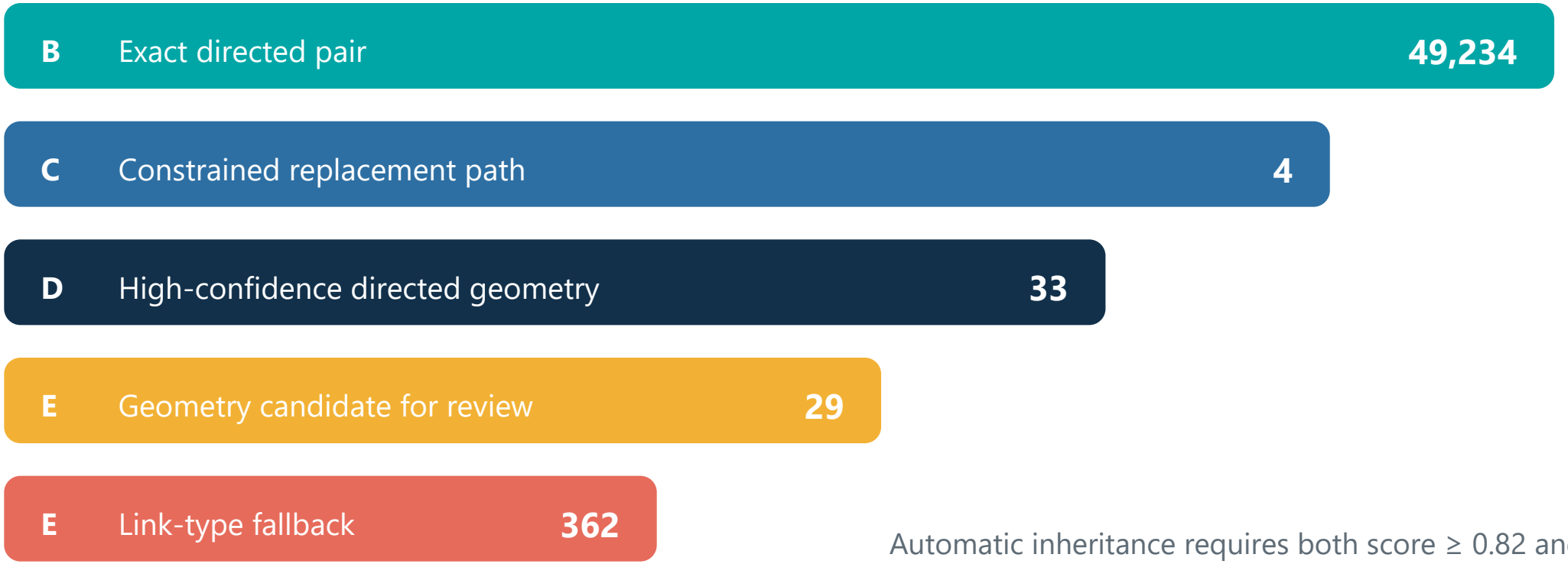
Other attributes

Leave empty when no exact lineage exists

Outcome 49,234 exact transfers plus 428 documented link-type fallbacks. The generated dictionary covers all 49,662 future links and can be passed directly to assignment.

Method 2 adds evidence before fallback

Stronger evidence always wins. Borderline geometry is retained for review without changing the applied fallback



Automatic inheritance requires both score ≥ 0.82 and winner margin ≥ 0.15. Candidates below either gate keep the link-type QVDF fallback.

Hierarchy follows CUBE_Future_Network_Attribute_Transfer_Framework.pdf

The framework recovers 37 links automatically

It also exposes 29 plausible geometry candidates without silently applying them

Resolution of the 428 exception links



Path 4 | Geometry 33 | Review candidate 29 | Fallback 362

37
automatic lineage recoveries

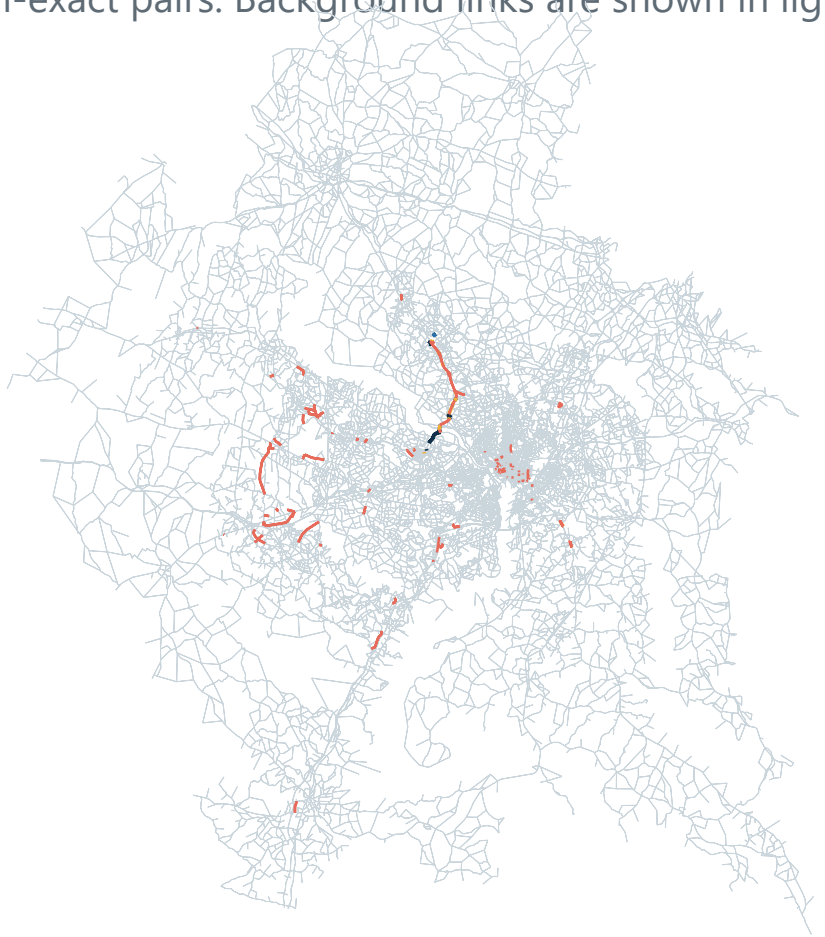
29
candidates held for manual review

391
links receiving QVDF fallback

The framework reduces fallback from 428 to 391 links, an 8.6% recovery within the exception set.

Exceptions cluster around future network edits

Regional view of the 428 non-exact pairs. Background links are shown in light gray



Replacement path 4

Accepted geometry 33

Review candidate 29

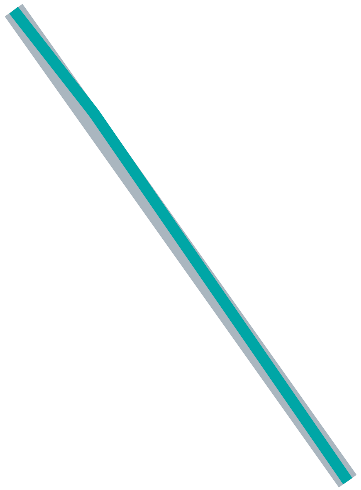
Link-type fallback 362

A real split link is recovered as a directed path

The 2025 pair 22417 to 22420 becomes two 2050 child links through node 23717

2025

22417 to 22420



2050

22417 to 23717 to 22420

Both directions are found. The four child links inherit the parent calibration with a score of 0.9999.

Attached links are excluded

Only links on the accepted directed replacement path inherit the parent values.

Geometry recovery is deliberately conservative

Candidate score and separation from the runner-up determine whether calibration is inherited



33
accepted geometry matches

29
candidates held at fallback

The held candidates remain in the crosswalk with their proposed base pair, score, and margin. They can be reviewed without rerunning the match.

The methods serve different governance needs

Decision	Simple	Framework
Automatic evidence	Exact pair only	Exact, path, high-confidence geometry
Inherited exceptions	0	37
Geometry review queue	None	29 candidates
QVDF fallback links	428	391
Non-QVDF behavior	Empty	Empty unless lineage is accepted
Best use	Immediate production	Governed recovery and future lineage

Both methods create a complete QVDF dictionary and preserve a row-level audit trail.

The implementation is ready for repeatable runs

One command creates the dictionary, crosswalk, missing-pair log, spatial layer, and summary

future_qvdf_node_pair_overrides.npy	Assignment-ready AM, MD, and PM QVDF dictionary
attribute_transfer_crosswalk.csv	One provenance row for every 2050 directed link
missing_2025_node_pairs.csv	The requested log of all 428 missing calibrated pairs
attribute_transfer_crosswalk.gpkg	GIS-ready spatial review layer
attribute_transfer_summary.json	Counts, thresholds, hashes, and source paths

```
python -m nvta_taplite_workflow transfer  --method simple|framework  --base-network ...  --future-network ...
```

Recommended operating decision

Use the deterministic method now. Keep the framework method as the governed recovery layer

Production default

Run the simple method for the 2050 assignment. It is transparent, complete, and never invents lineage.

Controlled enhancement

Use the framework result to review 29 geometry candidates and approve additional inheritance only where the evidence supports it.

Long-term fix

Add PARENT_BASE_UID during network editing. Explicit lineage will turn split-link transfer from inference into a deterministic operation.